

1. A college collected data about students based on **Study Hours** and **Attendance (%)**.

Student	Study Hours (x)	Attendance (y)	Result
S1	2	40	Fail
S2	4	45	Fail
S3	5	60	Pass
S4	6	65	Pass
S5	7	70	Pass
S6	3	50	Fail

A new student has: **Study Hours = 5 Attendance = 55** Using **K = 3**, classify whether the student will **Pass or Fail**.

2. A bank wants to approve loans based on: Income (in ₹10,000), Age, Credit Score

Person	Income	Age	Credit Score	Loan
P1	2	22	5	No
P2	3	25	6	No
P3	5	30	7	Yes
P4	6	35	8	Yes
P5	7	40	9	Yes
P6	4	28	6	No

New Applicant: **Income = 5 ,Age = 29 ,Credit Score = 7** Use **K = 3** and predict **Loan Approved or Not**.

3. The following dataset contains factors that identifies species of a tree
.Solve using KNN

Sepal Length	Sepal Width	Species
5.3	3.7	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	Versicolor

Sepal Length	Sepal Width	Species
5.2	3.1	?

4. A system predicts whether a leaf is **Healthy** or **Diseased** based on: Leaf Moisture (%), Temperature (°C)

Leaf	Moisture	Temp	Class
L1	80	25	Healthy
L2	82	26	Healthy
L3	78	24	Healthy
L4	60	32	Diseased
L5	58	33	Diseased
L6	62	31	Diseased

Test leaf: **Moisture = 75 ,Temp = 27** Using **K = 3**, classify the leaf.